BOOST UP PDFS | Quantitative Aptitude | Speed, Distance & Time (Easy Level Part-1)

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1. A train 200 m long running at 36 kmph takes 55 seconds to cross a bridge. The length of the bridge is

a. 375 m.

b. 300 m.

c. 350 m.

d. 325 m.

e. None of these

2. A train 250m long takes 30s to cross a man running at a speed of 6kmph in the direction opposite to that of train. What is the speed of the train?

a.20kmph

b.22kmph

c.23kmph

d.24kmph

e. None of these

3. A train 160m long passes a standing man in 25s. What is the speed of the train ?

a.21kmph

b.23kmph

c.25kmph

d.27kmph

e. None of these

4. A train 350m long takes 35s to cross a man running at a speed of 5kmph in the direction same to that of train. What is the speed of the train?

a.40kmph

b.41kmph

c.42kmph

d.43kmph

e. None of these

5. Without stopping the speed of the train is 120kmph , with stopping the speed of the train is 80kmph.Find the stop time of the train ?

a.10m

b.15m

c.20m

d.25m

e. None of these

6. Two trains approach each other at 20kmph and 24kmph from 2 places 240km apart. After how many hours they will meet ?

a.5hrs

b.5.25hrs

c.5.30hrs

d.5.45hrs

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e. None of these

7. A train 120m long passes an electric pole in 6s.How long will it take to cross a bridge of 240m long ?

a.18s

b.16m

c.13m

d.15s

e. None of these

8. A 120m long train is crosses the platform double its length in 30sec. What is the speed of the train ?

a.42kmph

b.43kmph

c.44kmph

d.40kmph

e. None of these

9.A 1km long wire is held by n poles, if 1 pole is removed the length of the gap is increased by 5/3m. What is the no of poles initially ?

a.24

b.26

c.23

d.22

e. None of these

10.P starts chasing a Q , 3hrs after the Q starts running. P takes 4hours to catch the Q. If the average speed of the P is 35kmph, Find the average speed of the Q ?

a.30kmph

b.16kmpg

c.20kmph

d.24kmph

e. None of these

11.Nakiren travelled 1200km by air which formed 2/5 of his trip. One third of the whole trip, he travelled by train and the rest of the journey by car. Find the distance travelled by car ?

a.850km

b.800km

c.700km

d.650km

e. None of these

12.A 180 m long train crosses another 270 m long train running in the opposite direction in 10.8 seconds. If the shorter train crosses a pole in 12 seconds, what is the speed of longer train?

a. 98 km/hr

b. 96 km/hr

c. 90 km/hr

d. 88 km/hr

e. 78 km/hr

13. Train – A crosses a stationary train – B in 50 seconds and a pole in 20 seconds with the same speed. The length of the train – A is 240 metres. What is the length of the stationary Train-B?

a. 360 metres

b. 260 metres

c. 300 metres

d. Cannot be determined

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e. None of these

14.Train A travelling at 52 km/hr crosses another train B, having three fourth of its length and travelling in opposite direction at 38 km/hr in 14 seconds. Train A passed a railway platform in 36 seconds. Find the length of platform.

a. 200 m

b. 300 m

c. 320 m

d. 400 m

e. 450 m

15.A train travels a distance of 600 km at a constant speed. If the speed of the train is increased by 5 km/hr, the journey would take 4 hrs less. Find the speed of the train.

a. 100 km/hr

b. 25 km/hr

c. 50 km/hr

d. 40 km/hr

e. 60 km/hr

16.If a car runs at 45km/hr, it reaches its destination late by 10 min but if runs at 60km/hr it is late by 4min. What is the correct time for the journey?

a. 24min

b. 14min

c. 32min

d. 20min

e. 46min

17.A bike rider starts at 40km/hr and he increases his speed in every 1 hour by 2km/hr. Then the maximum distance covered by him in 24 hrs is:

a. 682km

b. 540km

c. 620km

d. 612km

e. 500km

18.A man takes 4hrs 30min in walking to a certain place and riding back. He would have gained 2hrs by riding both ways. The time he would take to walk both ways, is:

a. 5hrs10min

b. 4hrs30min

c. 7hrs

d. 5hrs40min

e. 4hrs

19.Two trains of equal lengths take 10 seconds and 15 seconds respectively to cross a telegraph post. If the length of each train be 300 metres, in what time will they cross each other travelling in opposite direction?

a. 25 sec

b. 18 sec

c. 14 sec

d. 12 sec

e. 20 sec

20.A person sees a train passing over 3km long bridge. The length of the train is half that of bridge if

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the train clears the bridge in 4mins. What is the speed of the train ?

a. 67.5 km/hr

b. 50 km/hr

c. 62 km/hr

d. 48 km/hr

e. None of these

21.Two trains are running at 60 km/hr and 40 km/hr respectively in the same direction. Faster train completely passes a man sitting in the slower train in 54 seconds. What is the length of the fast train?

a. 100 m

b. 150 m

c. 300 m

d. 200 m

e. 250 m

22.A truck running from a city at a speed of 40 km/hr and the speed of the truck was increased by 2 km/hr at the end of every hour. Find the total distance covered by the truck in the first 5 hours of the journey.

a. 180 km

b. 120 km

c. 220 km

d. 90 km

e. 105 km

23.Two persons start running simultaneously around a rectangular track of length 700 m from the same point at speeds of 45 km/hr and 25 km/hr. When will they meet for the first time any where on the track if they are moving in opposite directions?

a. 50 sec.

b. 42 sec.

c. 25 sec.

d. 36 sec.

e. 40 sec.

24.I walk a certain distance and ride back taking a total time of 37 minutes. I could walk both ways in 55 minutes. How long would it take me to ride both ways?

a. 5 min.

b. 10 min.

c. 13 min.

d. 19 min.

e. None of these

25.A motor-cycle covers 40 km with a speed of 20 km/hr. find the speed of the motor-cycle for the next 40 km journey so that the average speed of the whole journey will be 30 km/hr.

a. 70 km/hr

b. 52.5 km/hr

c. 60 km/hr

d. 60.5 km/hr

e. None of these

26.Vikcy can row a certain distance downstream in 14 hoursand return the same distance in 21 hours. If the speed of the stream is 6 kmph, Find the speed of Vicky in the still water?

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a. 21 kmph
b. 15 kmph
c. 30 kmph
d. 35 kmph
e. None of these

27. Rahul can row a certain distance downstream in 12 hour and return the same distance in 18 hour. If the speed of Rahul in still water is 12 kmph, find the speed of the stream?

a. 2.1 kmph
b. 1.5 kmph
c. 4.4 kmph
d. 2.4 kmph
e. None of these

28. A Boat takes 128 min less to travel to 48 Km downstream than to travel the same distance upstream. If the speed of the stream is 3 Km/hr. Then Speed of Boat in still water is?

a. 6 Km/hr
b. 9 Km/hr
c. 12 Km/hr
d. 15 Km/hr
e. None

29.Sharma can row 36 km upstream in 4 hours. If the speed of the stream is 3 kmph, then find how far he can go downstream in 6 hours.

a. 68 km
b. 72 km
c. 70 km
d. 65 km
e. 75 km

30.A man can swim a certain distance downstream in 2 hours and return in 7 hours. If the rate of stream is 5 kmph then what is the speed of man in still water?

a. 12 kmph
b. 10 kmph
c. 6 kmph
d. 8 kmph
e. 9 kmph

31.A man swims downstream from one point to another which is 6 km apart in 1.5 hours. It covers the same distance upstream in 2 hours. Find the speed of the man in still water.

a. 4 kmph
b. 4.1 kmph
c. 4.2 kmph
d. 3.5 kmph
e. 4.5 kmph

32. A boat takes 8 hours to cover a distance while travelling upstream. Whereas while travelling downstream, it takes 6 hours, If the speed of the stream is 4 kmph, what is the speed of the boat in still water?

a. 16 kmph
b. 18 kmph
c. 28 kmph
d. Cannot be determined
e. None of these

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33. A man takes to row 80 km along the stream is half of the time taken to row against the stream. Find the speed of the boat in still water if the speed of stream is 6 km/hr?

a. 24 km/hr

b. 20 km/hr

c. 18 km/hr

d. 12 km/hr

e. None of these

34. A boat running downstream covers a distance of 40 km in 5 hrs and for covering the same distance upstream, it takes 10 hrs. What is the speed of the stream?

a. 5 km/hr

b. 2 km/hr

c. 6 km/hr

d. 4 km/hr

e. 3 km/hr

35.A Boat takes 128 min less to travel to 48 Km downstream than to travel the same distance upstream. If the speed of the stream is 3 Km/hr. Then Speed of Boat in still water is?

a. 6 Km/hr

b. 9 Km/hr

c. 12 Km/hr

d. 15 Km/hr

e. None of these

36.A man takes thrice as long to row a distance against the stream as to row the same distance in favour of the stream. The ratio of the speed of the boat (in still water) and the stream is:

a. 2:1

b. 3:2

c. 1:2

d. 2:3

e. None of these

37.A boat travels downstream for 14km and upstream for 9km. If the boat took total of 5 hours for its journey. What is the speed of the river flow if the speed of the boat in still water is 5km/hr?

a. 8km/hr

b. 2km/hr

c. 6km/hr

d. 5km/hr

e. 3km/hr

38.Rajiv can swim at 10 km/hr in still water. The river flows at 3 km/hr and it takes 12 hours more upstream than downstream for the same distance. How far is the place?

a. 182 km

b. 164 km

c. 176 km

d. 160 km

e. 175 km

39.Mr. Suresh can row to a place 48 km away and come back in 14 hours. He finds that he can row 4 km with the stream in the same time as 3 km against the stream. The rate of the stream is?

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a. 1 kmph
b. 3 kmph
c. 4 kmph
d. 6 kmph
e. None of these

40.A Boat going upstream takes 8 hours 24 minutes to cover a certain distance, while it takes 5 hours to cover 5/7 of the same distance running downstream. Then what is the ratio of the speed of boat to speed of water current?

a. 6:5
b. 11:5
c. 11:6
d. 11:1
e. 11:10

41.Mr.Ramesh's speed with the current is 20 kmph and the speed of the current is 5 kmph. Ramesh's speed against the current is?

a. 15 kmph
b. 19 kmph
c. 17 kmph
d. 10 kmph
e. None of these

42.A Boat takes total 16 hours for traveling downstream from point A to point B and coming back point C which is somewhere between A and B. If the speed of the Boat in Still water is 9 Km/hr and rate of stream is 6 Km/hr, then what is the distance between A and C?

a. 30 Km
b. 60 Km
c. 90 Km
d. 100 Km
e. Cannot be determined

43.Ravi can row 12 kmph in still water when the river is running at 6 kmph it takes him 1 hour to row to a place and to come back. How far is the place?

a. 5.5 km
b. 4.5 km
c. 8.2 km
d. 4.2 km
e. None of these

44.The different between downstream speed and upstream speed is 2 kmph and the total time taken during upstream and downstream is 2 hours. What is the upstream speed, if the downstream and upstream distance are 2 km each?

a. 5.2 kmph
b. 3.7 kmph
c. 2.8 kmph
d. 1.4 kmph
e. None of these

45.A Boat takes 128 min less to travel to 48 Km downstream than to travel the same distance upstream. If the speed of the stream is 3 Km/hr. Then Speed of Boat in still water is?

a. 6 Km/hr b. 9 Km/hr c. 12 Km/hr d. 15 Km/hr
e. None

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46.A Boat took 8 hours less to travel a distance downstream than to travel the same distance upstream. If the speed of a boat in still water is 9 Km/hr and speed of a stream is 3 Km/hr. In total how much distance traveled by boat?

a. 96 Km

b. 144 Km

c. 164 Km

d. 192 Km

e. 216 Km

47.A boat running downstream covers 28 kms in 7 hours, while for covering the same distance upstream it takes 4 hours. What is the speed of the boat in still water and ?

a. 4km/hr

b. 4.5km/hr

c. 5km/hr

d. 5.5km/hr

e. None

48.In a stream, B lies in between A and C such that it is equidistant from both A and C. A boat can go from A to B and back in 6 h 30 minutes while it goes from A to C in 9 h. How long would it take to go from C to A?

a. 2 h

b. 4.5 h

c. 4 h

d. 3.5 h

e. None of these

49.A man can row 8 km/hr in still water. If it takes him thrice as long to row up, as to row down the river, then the rate of current in the stream would be

a. 6.5km/hr

b. 4.5km/hr

c. 4km/hr

d. 5.5km/hr

e. None

50.A boatman goes 4 km against the current of the stream in 1 hour and goes 2 km along the current in 10 minutes. How long will it take to go 24 km in stationary water?

a. 5hrs

b. 3hrs

c. 2hrs

d. 6hrs

e. None

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Answer Key with Detailed Solution

1. C
Speed of train = 36 kmph
= 36 × (5 / 18) = 10 m / sec
If the length of bridge be x metre, then
10 = (200 + x) / 55
200 + x = 550
x = 550 – 200 = 350 metre

2. D
Speed = x
Relative speed = (x+6)kmph
(x+6) = (250×18)/(30×5)
(x+6) = (4500)/(150)
X+6 = 30
X = 30 – 6 = 24kmph

3. B
Speed = (160×18)/(25×5)
= 2880/125 = 23.04 = 23kmph

4. B
Speed = x
Relative speed = (x-5)kmph
(x-5) = (350×18)/(35×5)
(x-5) = (6300)/(175)
X-5 = 36
X = 36 + 5 = 41kmph

5. C
Stopping time = (120 – 80)/ 120 = 40/120 =1/3h
= 60/3 = 20 min

6. D
20x + 24x = 240
44x = 240
X = 240/44 = 5.45hrs

7. A
Speed = 120/6 = 20m/s
Time taken to cross the bridge = (120+240)/20
= 360/20
= 18s

8. B
120+240 / 30 =12
12*18/5 = 43.2 = 43kmph

9. B
[1000/(n-2)] [1000/(n-1)] = 5/3
n =26

10. C
Distance P covers = 35*4 = 140
After 3hrs= 4+3 = 7
Avg speed of Q = 140/7 = 20kmph

11. B
2x/5 = 1200……air
X = 3000km……total distance
3000/3 = 1000km……train
Train = 3000 – (1200+1000) = 800km(car)

12. B
Sol. Speed of shorter train $= \frac{180}{12} = 15\ \text{m/sec}$
ATQ,
$(15 + x) = \frac{180 + 270}{10.8}$, where x = speed of longer train
$\Rightarrow 15 + x = \frac{4500}{108}$
$\Rightarrow x = \frac{4500}{108} - 15$
$\Rightarrow x = \frac{2880}{108} \times \frac{18}{5} = 96$ kmph

13. A
Sol. Let length of train-B is l metres
$\frac{240 + l}{50} = \frac{240}{20}$
$\Rightarrow l = 360$ metres

14. C

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Sol. Let length of first train is x metre

Length of second train = $\frac{3x}{4}$

$$\therefore \frac{x+\frac{3x}{4}}{(52+38)\times\frac{5}{18}} = 14$$

$$\Rightarrow \frac{7x}{100} = 14 \Rightarrow x = 200 \text{ m}$$

∴ Length of platform

$$= 36 \times 52 \times \frac{5}{18} - 200$$

$$= 520 - 200 = 320 \text{ m}$$

15. B

Sol. Using formula, $\frac{S(S+5)}{5} \times 4 = 600$

$$S(S+5) = 750 = 25\ (25+5)$$

Speed of the train = 25 km/hr

Alternately,

$$\frac{600}{s} - \frac{600}{s+5} = 4$$

$$\frac{600s+3000-600s}{s(s+5)} = 4$$

$$s = 25\ km/hr$$

16. B

Distance = diff in time *(S1*S2)/S1-S2
D=[10-4]/60hr * (45*60)/[60-45] = 6/60 * 45*60/15 ==>18km
time T=D/S(take any one of the speed. 18/45 =2/5hrs = 2/5*60=24min
then correct time is 24-10=14mins.

17. D

Speed of the rider: 40km/hr
He increases his speed in every 1 hr by 2km/hr.
Distance covered by every 1 hr will be 40,42,44….upto 12terms. i.e. (for 24hrs)
Sum of 1st n terms=n/2 ((2a+(n-1)d.
12/2 * (2*40+11*2)==>12/2 * (80+22)==>612km

18. C

W+R = 4hrs 30min ie 9/2hrs
R+R=2hrs==>R=1hr
then 2W=9/2-1=7/2 , W=7/2*2=7hrs

19. D

Speed of the first train = [300/10] m/sec = 30 m/sec.
Speed of the second train = [300/15] m/sec =20 m/sec.
speed = (30 + 20) m/sec = 50m/sec.
Required time = (300+300)/50secc = 12 sec.

20. A

Distance covered in 4/60 hours = (3 + 3/2) = 9/2 km
Distance covered in 1 hour = (9/2) x (60/4) = 67.5km/hr.
So, speed of the train = 67.5 km/hr

21. C

Relative speed = (60 – 40) km/hr = 20km/hr.
Length of the faster train = (20*54*5)/18 = 300m

22. C

The total distance covered by the truck in the first 5 hours = 40 + 42 + 44 + 46 + 48
sum of five terms in AP whose first term is 40 and last term is 48 = 5/2 [40 + 48] = 220 km.

23. D

Time taken to meet for the first time = length of the track / relative speed = 700 / (45 + 25)
= (700*18)/(70*5) = 36 sec.

24. D

Let the distance be x, then,

(Time taken to walk x km) + (Time taken to ride x km) = 37min.

⇒ (Time taken to walk 2x km) + (Time taken to ride 2x km) = 74 min.

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But, time taken to walk 2x km = 55 min.

So, Time to ride 2x km = (74 - 55) min ⇒ 19 min.

25. C

Average speed of whole journey = 30 km/hr

x = speed of first 40 km = 20 km/hr

$$\Rightarrow 30 = \frac{2 \times 20 \times y}{20 + y}$$

60

26. C

Speed of Vikcy in still water = x

Downstream Speed = (x + 6)

Upstream Speed = (x – 6)

Downstream Distance = Upstream Distance

14(x + 6) = 21(x – 6)

2x + 12 = 3x – 18➔x=30

27. D

Let the speed of the stream be x kmph

Down stream = (12+x)

Up stream = (12−x)

suppose the distance traveled be y

y/(12+x) = 12 —(1)

y/(12−x)= 18 —-(2)

From eqn (1) and (2)

x= 2.4 kmph

28. C

32/15 = 48(1/s-3 – 1/s+3)

s= 12

29. B

Speed of the stream = 3 kmph

Let the speed of Sharma be x kmph.

According to the question, 36 / (x – 3) = 4

x = 36/4 + 3 = 9 kmph

Distance travelled downstream in 6 hours

= 6 × (9 + 3) = 72 km.

30. E

Let the speed of the man in still water be x kmph.

Then, (x + 5) × 2 = (x – 5) × 7

or, 2x + 10 – 7x + 35 = 0

or, 5x = 45

or, x = 9.

31. D

Speed of man downstream = 6 / 1.5 kmph = 4 kmph

Speed of man upstream = 6 / 2 = 3 kmph

or, Speed of man = 1 / 2 × (4 +3) kmph

= 3.5 kmph.

32. C

Let the speed of the boat in still water be x kmph.

Then, (x + 4) ×6 = (x – 4) × 8

or, 6x + 24 = 8x – 32

or, x = 28 kmph.

33. C

Let us take the speed of the boat in still water be x km/hr,

80/(x+6) = 1/2*(80/(x-6))

2x – 12 = x + 6

X = 18

Speed of the boat in still water = 18 km/hr

34. B

Downstream speed = 40/5 = 8 km/hr

Upstream speed = 40/10 = 4 km/hr

So, speed of stream = 1/2*(8-4) = 4/2 = 2 km/hr

35. C

Let the speed of boat be s,

According to the question,

128/60 = 48/s-3 – 48/s+3

128/60 = 48(1/s-3 – 1/s+3)

135 = s^2 +3s -3s – 9

s^2 = 144

s= 12

The speed of boat in still water = 12 km/hr

36. A

Speed of downstream : Speed of upstream = 3 : 1(3x, x)

The ratio of speed of boat in still water and speed of stream is,

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= > (3x+x)/2 : (3x-x)/2
= > 4x/2 : 2x/2
= >2 : 1

37. B

Let the speed of the stream be x km/hr.
Upward speed = (5 – x)km/hr.
Downward speed = (5 + x)km/hr
According to the question,
= > 14/(5+x) + 9/(5-x) = 5
=> (70 – 14x + 45 + 9x) / [(5+x) (5 –x)] = 5
= > 115 – 5x = 125 – 5x2
= > 5x2 – 5x – 10 = 0
= > x2 – x – 2 = 0
= > x = 2, -1(eliminate.
=> x = 2 km/hr
Speed of stream = 2 km/hr

38. A

x/(10-3) – x/(10+3) =12
x/7 – x/13 =12
13x-7x / 91 =12
= > X= 182 km

39. A

Downstream speed = 4/x kmph
upstream speed = 3/x kmph
48/(4/x) + 48/(3/x) = 14
Solving we get x = 1/2 kmph
So, Speed of downstream = 8 kmph, Speed of upstream = 6 kmph
Stream Speed = 1/2(8 – 6) kmph = 1 kmph

40. D

(S-R)*42/5 = (S+R)*7
S:R = 11:1

41. D

Ramesh's speed with the current = 20 kmph
=> Ramesh's speed + speed of the current = 20 kmph
Speed of the current = 5 kmph
Speed of Ramesh = 20 – 5 = 15 kmph
Ramesh's speed against the current = speed of Ramesh – speed of the current = 15 – 5 = 10 kmph

42. E

16 = D/9+6 + x/9-6

43. B

Downstream Speed = 18 kmph
Upstream Speed = 6 kmph
Distance = x
x/18 + x/6 = 1
18x + 6x = 108
24x = 108
x = 4.5 km

44. D

2/x + 2/(x+2) = 2.
$x^2 – 2 = 0$
x = 1.414kmph

45. C

32/15 = 48(1/s-3 – 1/s+3)
s= 12

46. D

8 =D(1/6 – 1/12)
D =96
Total = 96+96 =192

47. D

Downstream=28/7=4km/hr
Upstream=28/4=7km/hr
Speed of the boat u=(4+7) /2
=11/2=5.5km/hr.

48. C

$$\text{Time taken to go from A to B} = \frac{\text{Time taken to go from A to C}}{2}$$
$$= \frac{9}{2}$$
= 4 h 30 min
Time taken to go from B to A = 6.5 – 4.5 = 2h.
Time taken to go from C to A = 2 × 2 = 4 h

49. C

Ratio 3:1
u=8km/hr

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(8-v)*3t = (8+v)*t
24-3v=8+v
4v=16
v=4km/hr.

50. B

Upstream=4/1=4km/hr
Downstream=2*60/10=12km/hr
Speed in still water=(12+4)/2
=16/2=8km/hr.
For 24 km time=24/8
=3hrs.

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